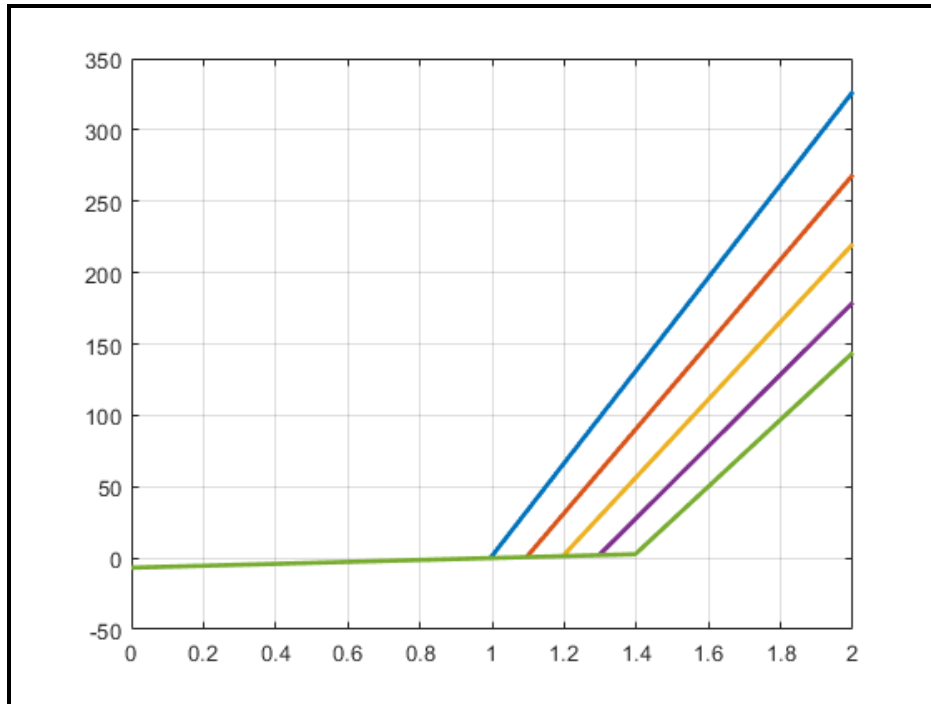


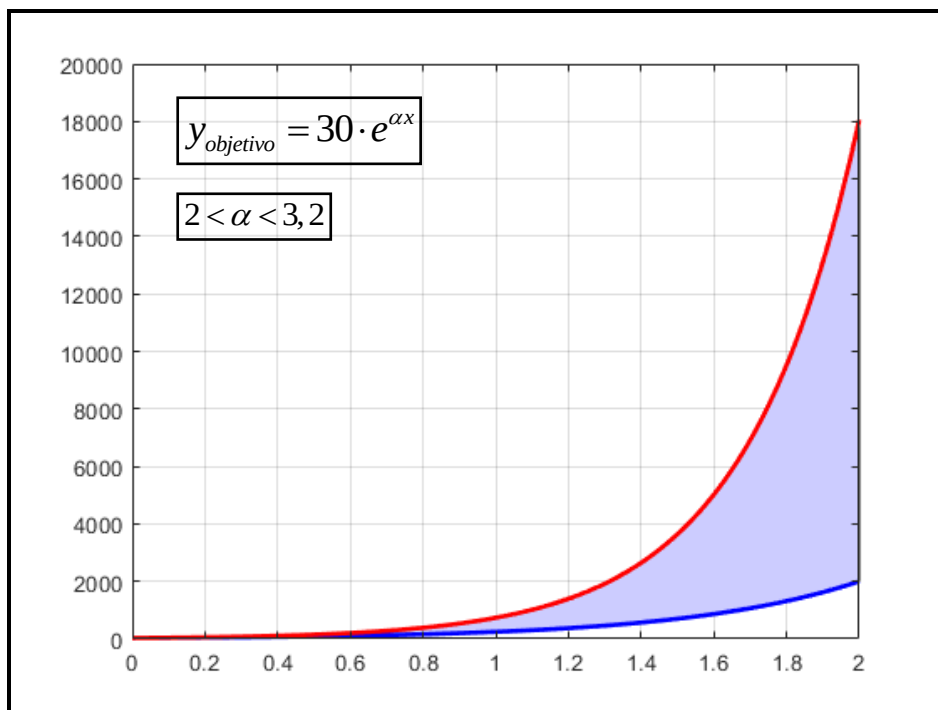
Aproximación de la función objetivo con una combinación lineal de coeficientes enteros para 5 funciones base, realizado con una Red Neuronal entrenada.

Los datos de entrenamiento se logran buscando los valores óptimos con el método de los Multiplicadores de Lagrange (reales). Luego se entrenó a una Red Neuronal con datos generados por el método numérico (reales). Luego se compararon las predicciones del método numérico y los de la red neuronal (reales). Luego se redondearon las predicciones de la red neuronal (enteros), y se generaron las curvas de aproximación finales.

Funciones Base



Rango de variación de las funciones objetivo para aproximaciones con $R^2 > 0,94$



Los algoritmos de aproximación numérica y de red neuronal se realizaron con números reales **R**

TABLA 1: COEFICIENTES DE LAGRANGE (REALES)

α	c1	c2	c3	c4	c5	R ²
2.00	3.7414	0.0000	0.0000	0.0000	1.6296	0.93460524
2.20	4.6846	0.0000	0.0000	0.0000	4.3144	0.95104031
2.50	6.4478	0.0000	0.0000	0.0000	12.1093	0.96340625
2.70	8.3497	0.0000	0.0000	0.0000	20.0000	0.96640918
2.75	8.4921	0.0000	0.0000	2.7970	20.0000	0.96587076
2.80	8.5634	0.0000	0.0000	6.0488	20.0000	0.96518202
3.00	9.7030	0.0000	0.5304	20.0000	20.0000	0.96087980
3.20	4.5380	5.9302	20.0000	20.0000	20.0000	0.95066264

TABLA 2: COEFICIENTES DE LA RN (REALES)

α	c1	c2	c3	c4	c5	R ²
2.00	3.7118	0.0417	-0.0018	0.0200	1.6842	0.93437426
2.20	4.6367	0.0292	-0.0739	0.0007	4.1766	0.95048381
2.50	6.4222	0.0106	-0.1291	-0.0076	11.9841	0.96320810
2.70	8.3040	-0.0005	-0.1873	0.1312	20.0717	0.96643538
2.75	8.5139	-0.0187	-0.3098	2.5823	20.1547	0.96570335
2.80	8.5898	-0.0165	-0.2938	5.9994	20.0835	0.96515543
3.00	9.2452	0.0075	1.6371	19.7878	20.0622	0.96080772
3.20	5.5987	4.6141	19.6302	20.6896	19.5448	0.95058217

TABLA 3: COEFICIENTES DE LA RN (REDONDEADOS A ENTEROS)

α	c1	c2	c3	c4	c5	R ²
2.00	4	0	0	0	2	0.91974463
2.20	5	0	0	0	4	0.94884288
2.50	6	0	0	0	12	0.95997129
2.70	8	0	0	0	20	0.96557706
2.75	9	0	0	3	20	0.96385530
2.80	9	0	0	6	20	0.96438468
3.00	9	0	2	20	20	0.96076416
3.20	6	5	20	21	20	0.94957728

RESUMEN DE R² - COMPARATIVA

α	R ² Lagrange	R ² RN real	R ² RN redondeada
2.00	0.93460524	0.93437426	0.91974463
2.20	0.95104031	0.95048381	0.94884288
2.50	0.96340625	0.96320810	0.95997129
2.70	0.96640918	0.96643538	0.96557706
2.75	0.96587076	0.96570335	0.96385530
2.80	0.96518202	0.96515543	0.96438468
3.00	0.96087980	0.96080772	0.96076416
3.20	0.95066264	0.95058217	0.94957728

ESTADÍSTICAS DE R²

Método	Mínimo	Máximo	Promedio
Lagrange	0.93460524	0.96640918	0.95725702
RN real	0.93437426	0.96643538	0.95709378
RN redondeada	0.91974463	0.96557706	0.95408966

La función aproximada será del siguiente tipo:

$$y_{aprox}(x) = c_1 \cdot b_1(x) + c_2 \cdot b_2(x) + c_3 \cdot b_3(x) + c_4 \cdot b_4(x) + c_5 \cdot b_5(x)$$

Donde los c_i son los coeficientes dados por los diferentes métodos de aproximación para cada valor de α correspondiente de la la función objetivo, y donde los $b_i(x)$ son las distintas curvas base.

Arquitectura de la Red Neuronal

```
model = Sequential([
    Dense(64, activation='relu', input_shape=(1,)), # Capa 1: Entrada → 64 neuronas
    Dense(128, activation='relu'), # Capa 2: 64 → 128 neuronas
    Dense(64, activation='relu'), # Capa 3: 128 → 64 neuronas
    Dense(5, activation='linear') # Capa 4 (salida): 64 → 5 neuronas
])
```

Arquitectura de la Red Neuronal (FCNN)

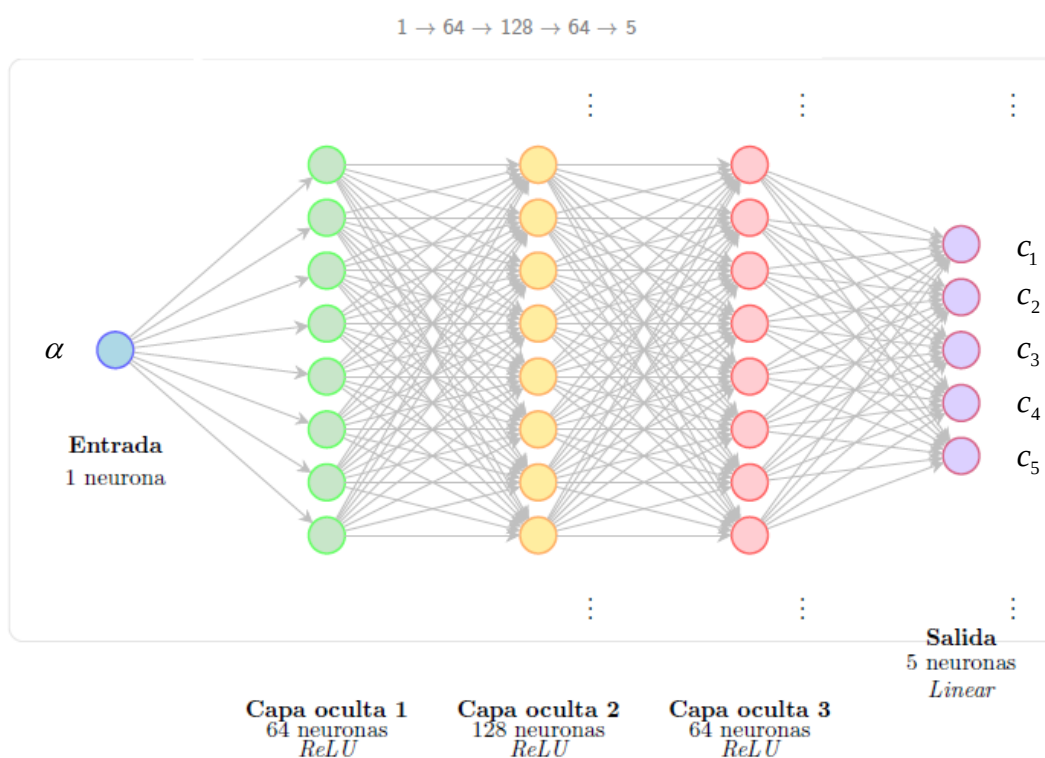
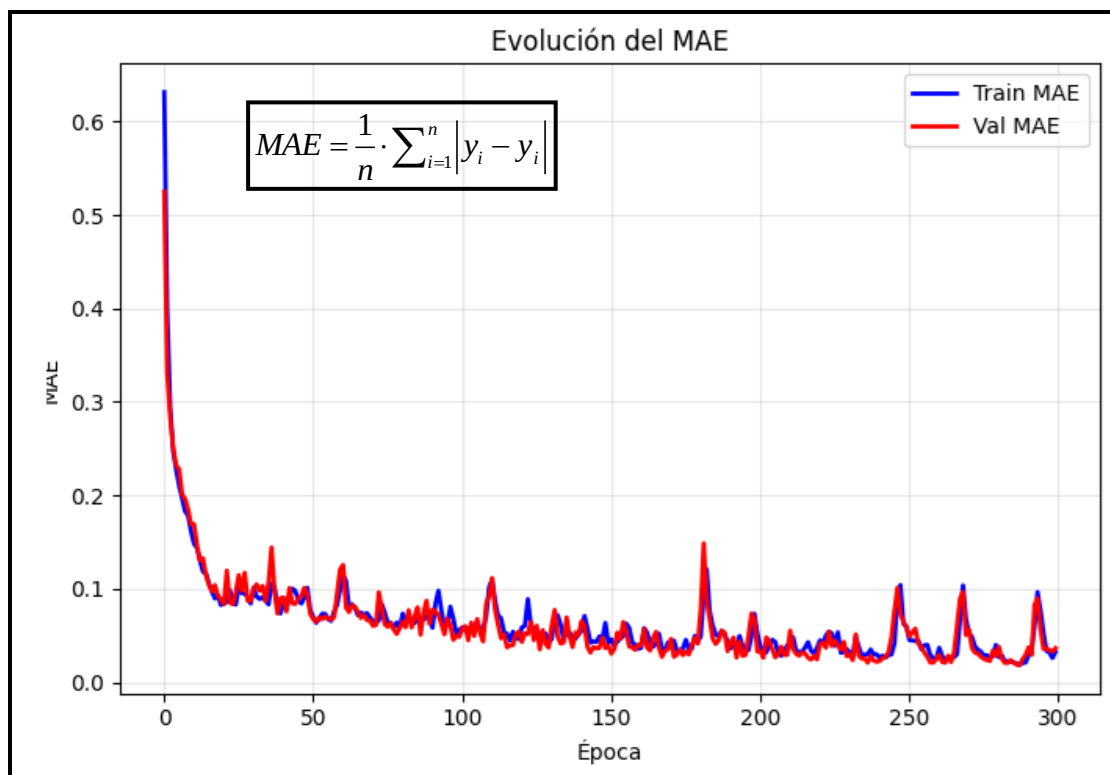
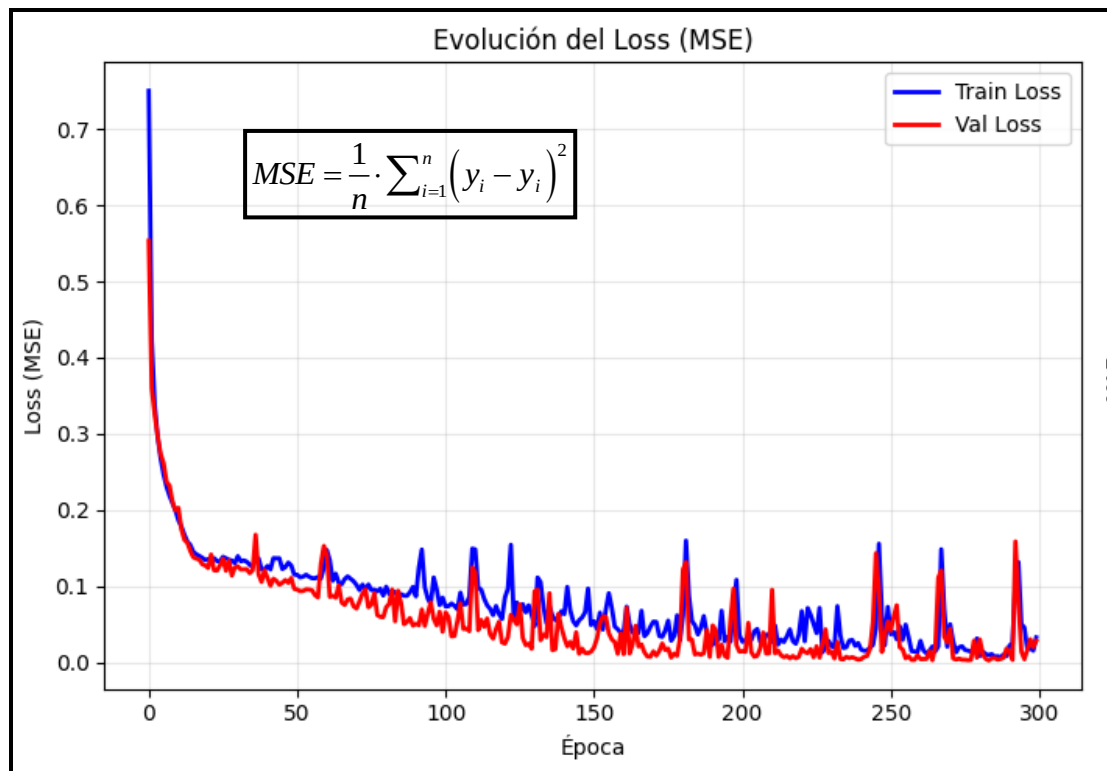
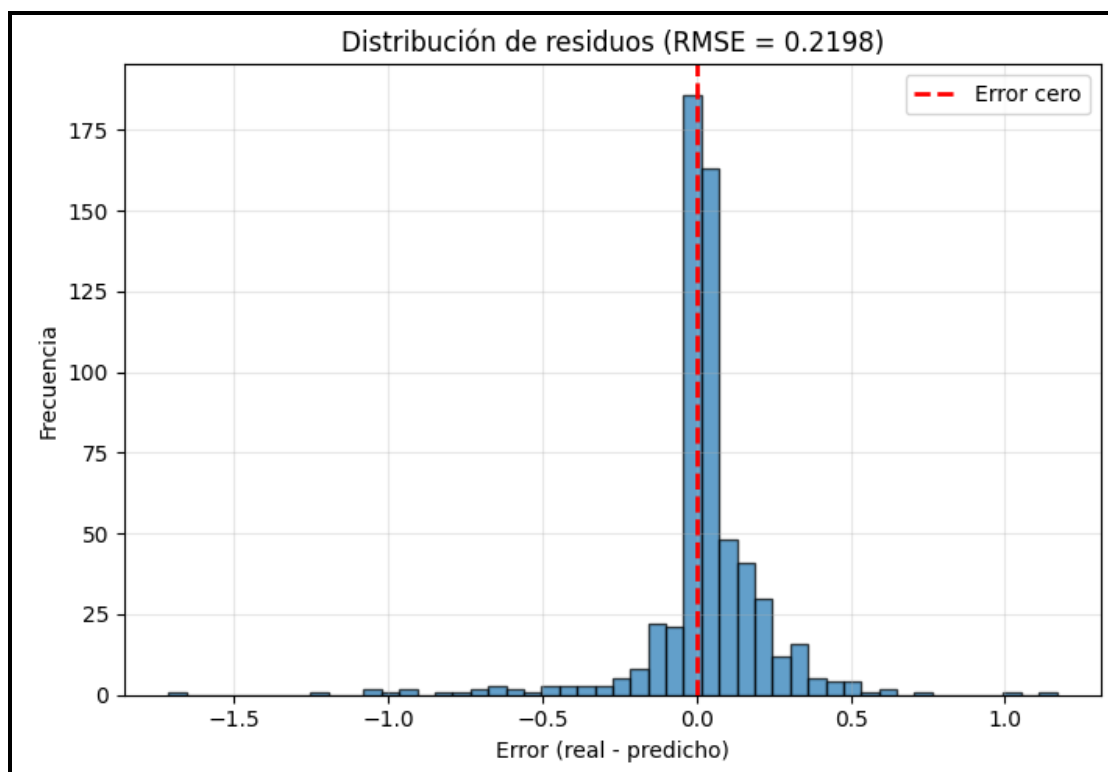
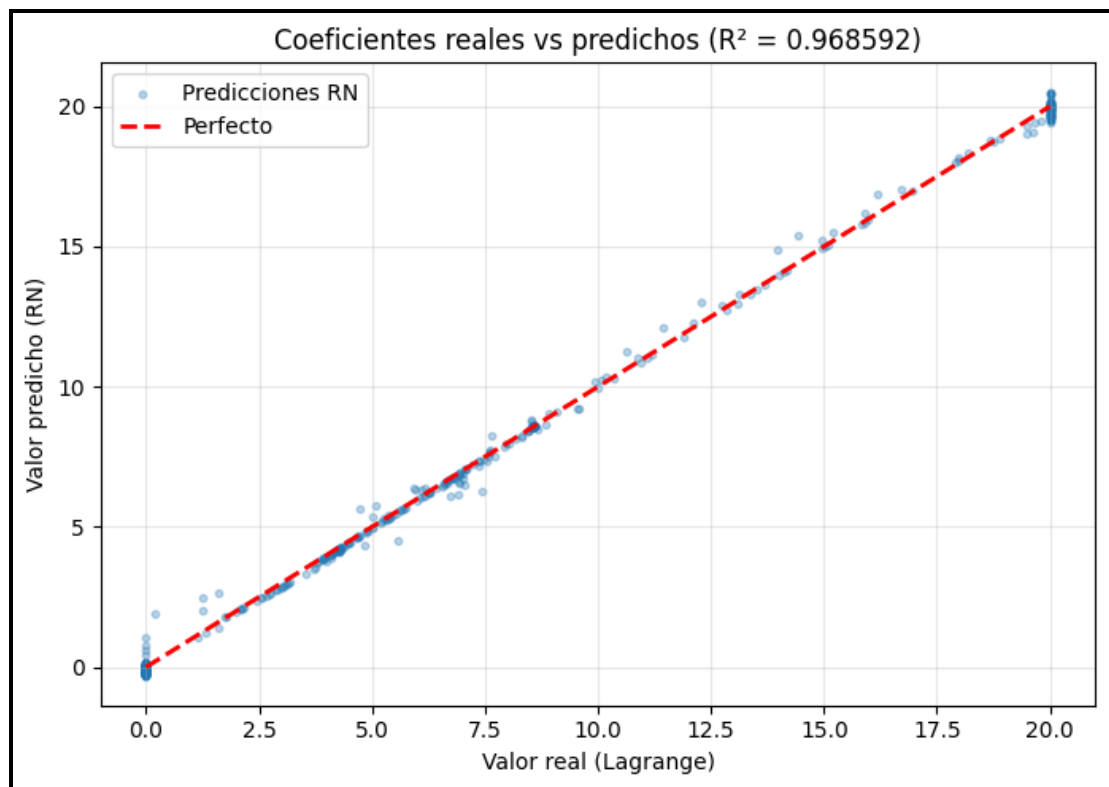


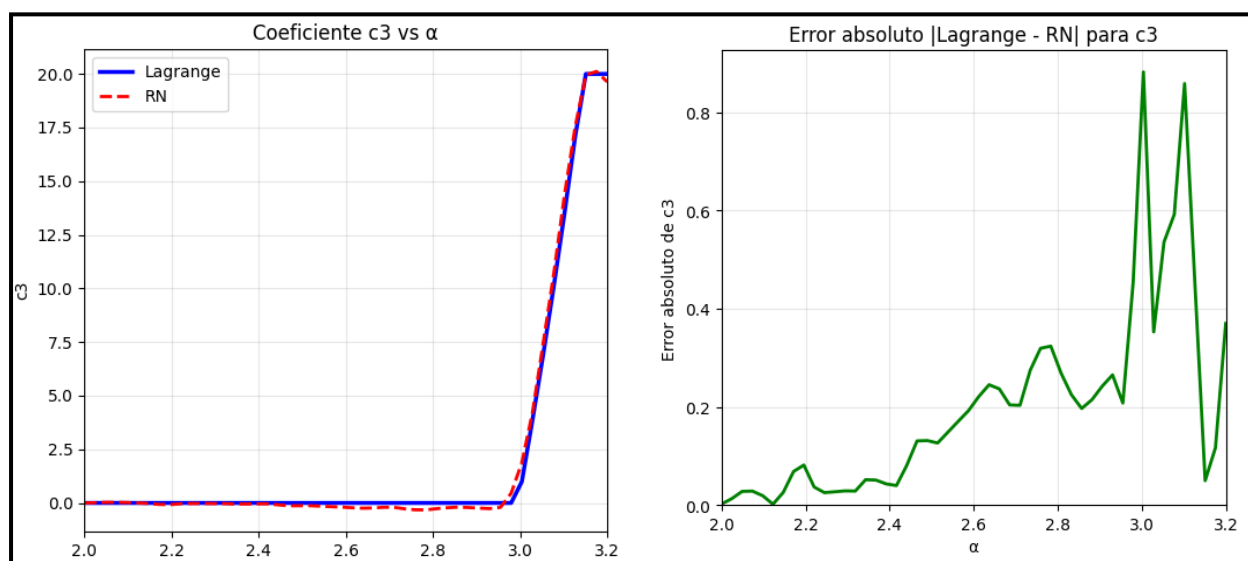
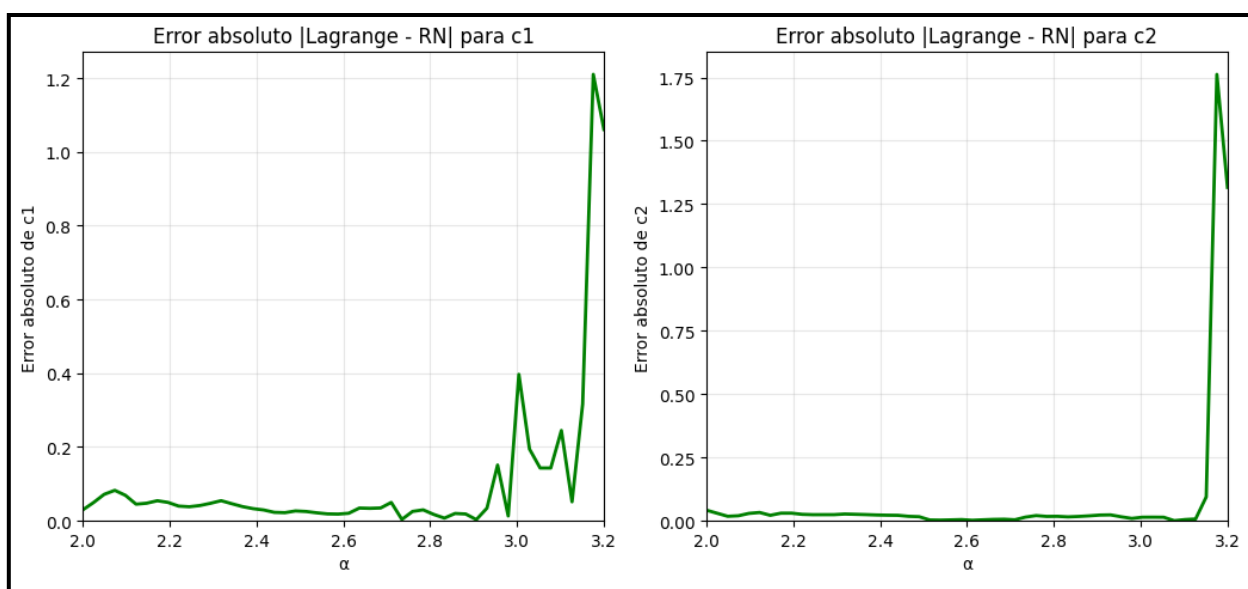
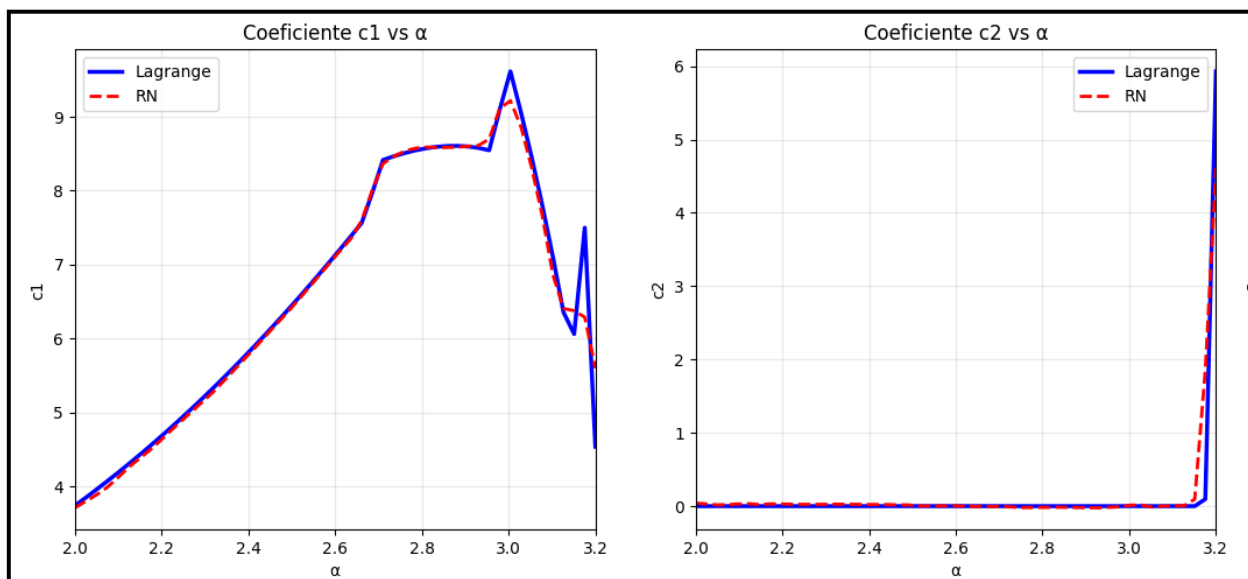
Figure 1: Arquitectura de la Red Neuronal completamente conectada (FCNN)
Se muestran 8 neuronas representativas por capa oculta.

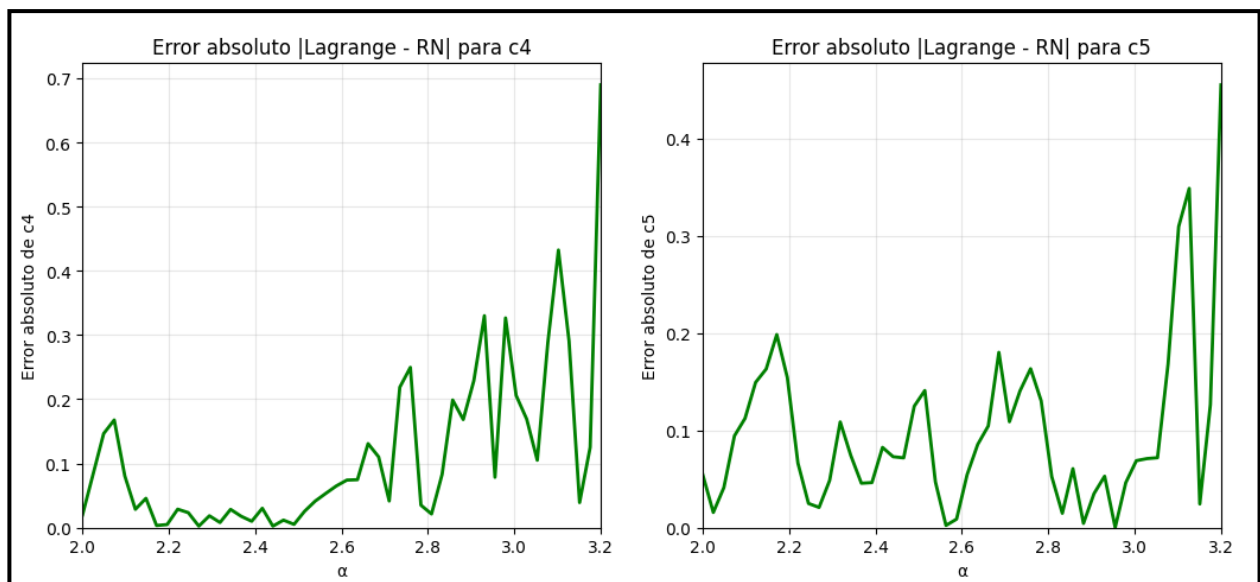
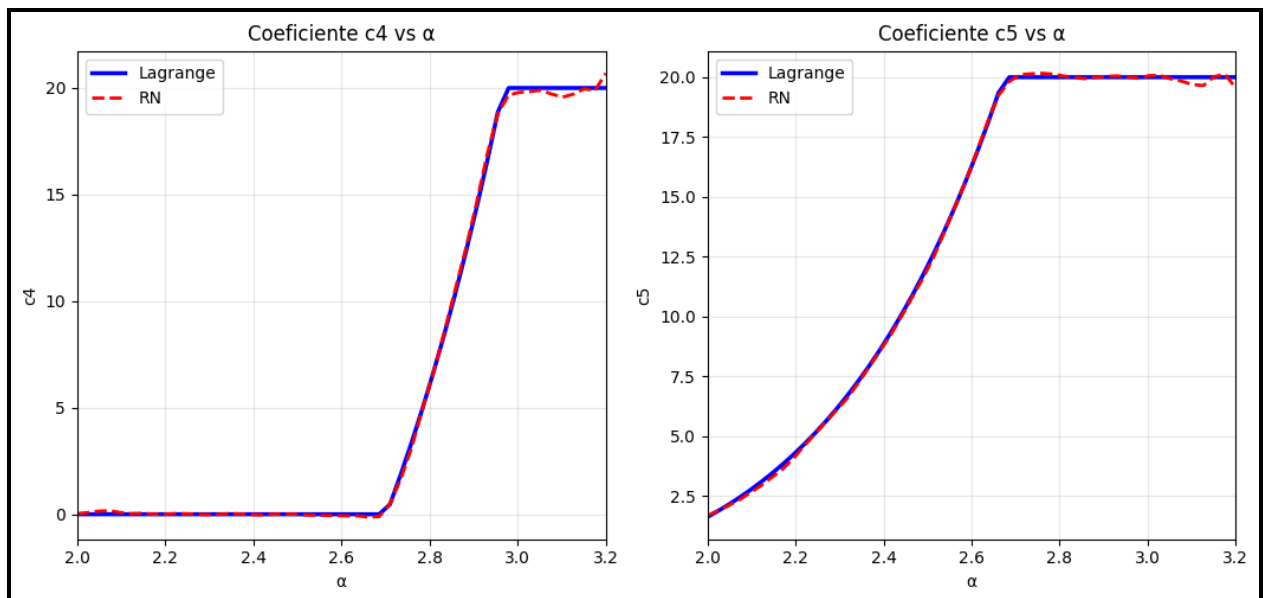
Métricas del entrenamiento de la Red Neuronal

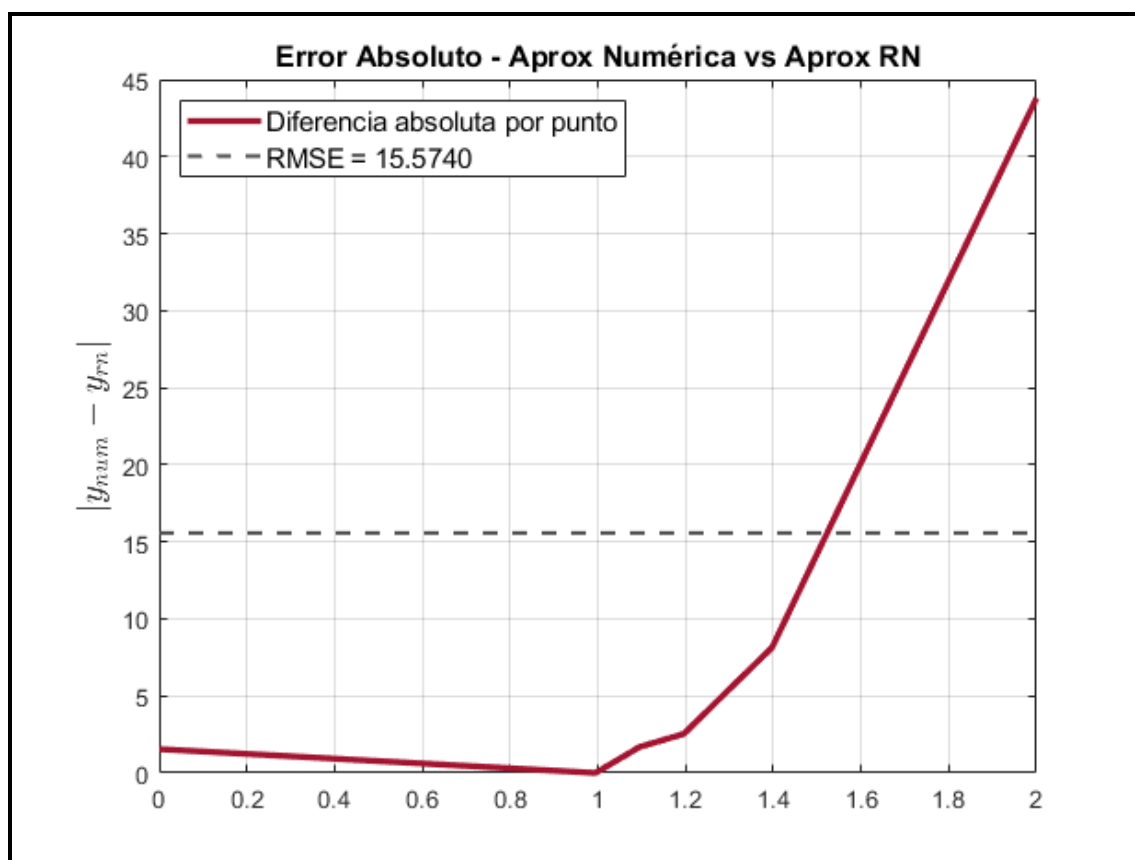
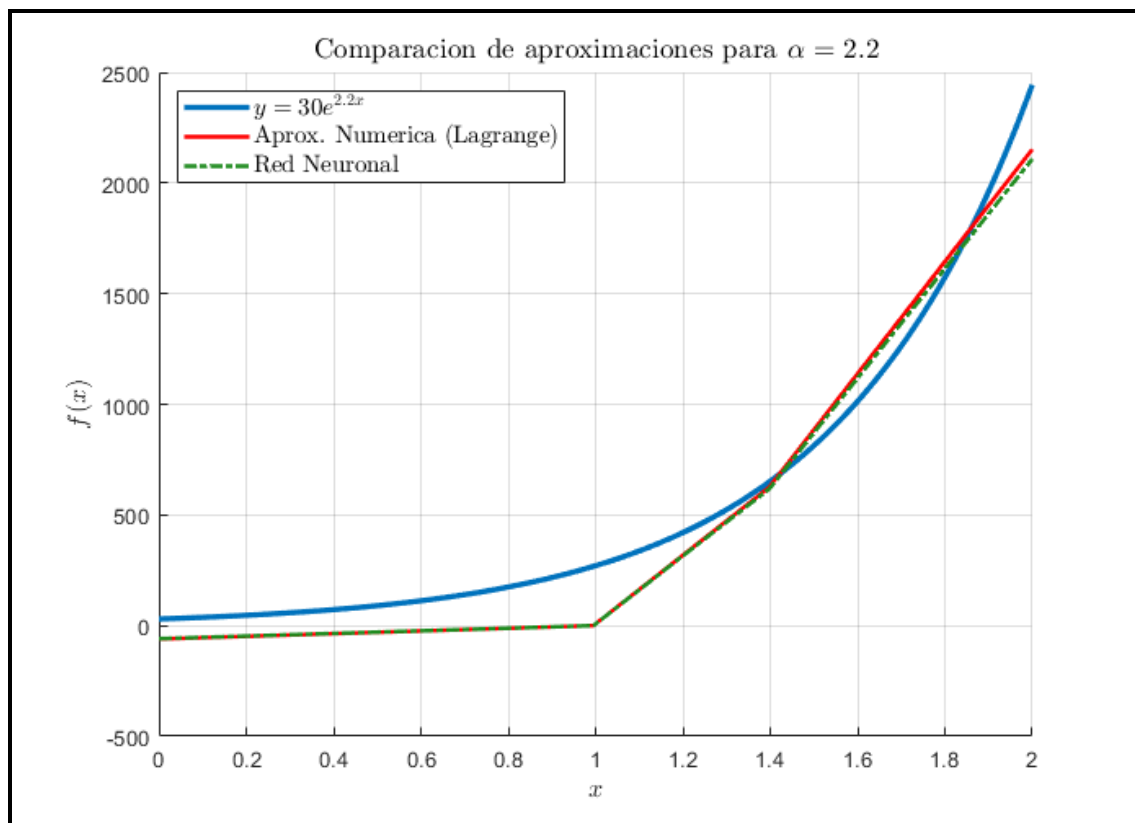
No hay overfitting ya que las pérdidas en el entrenamiento y el test disminuyen

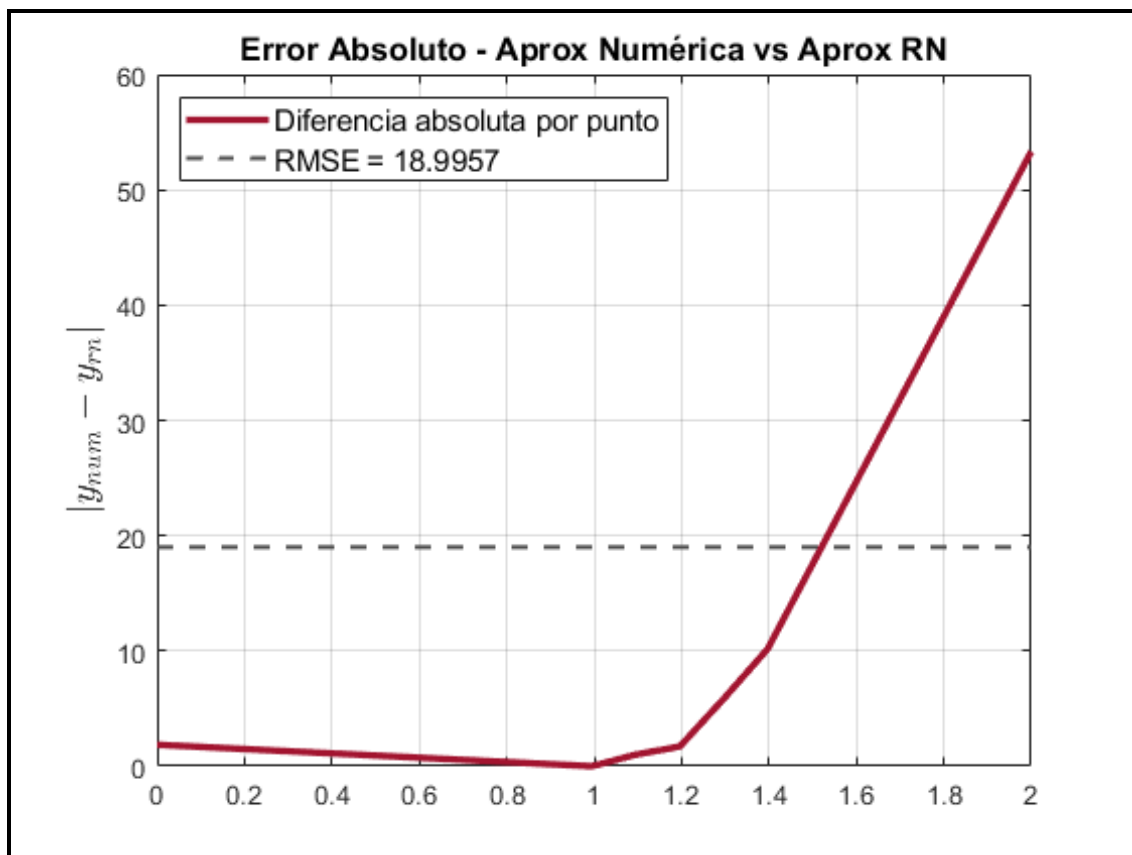
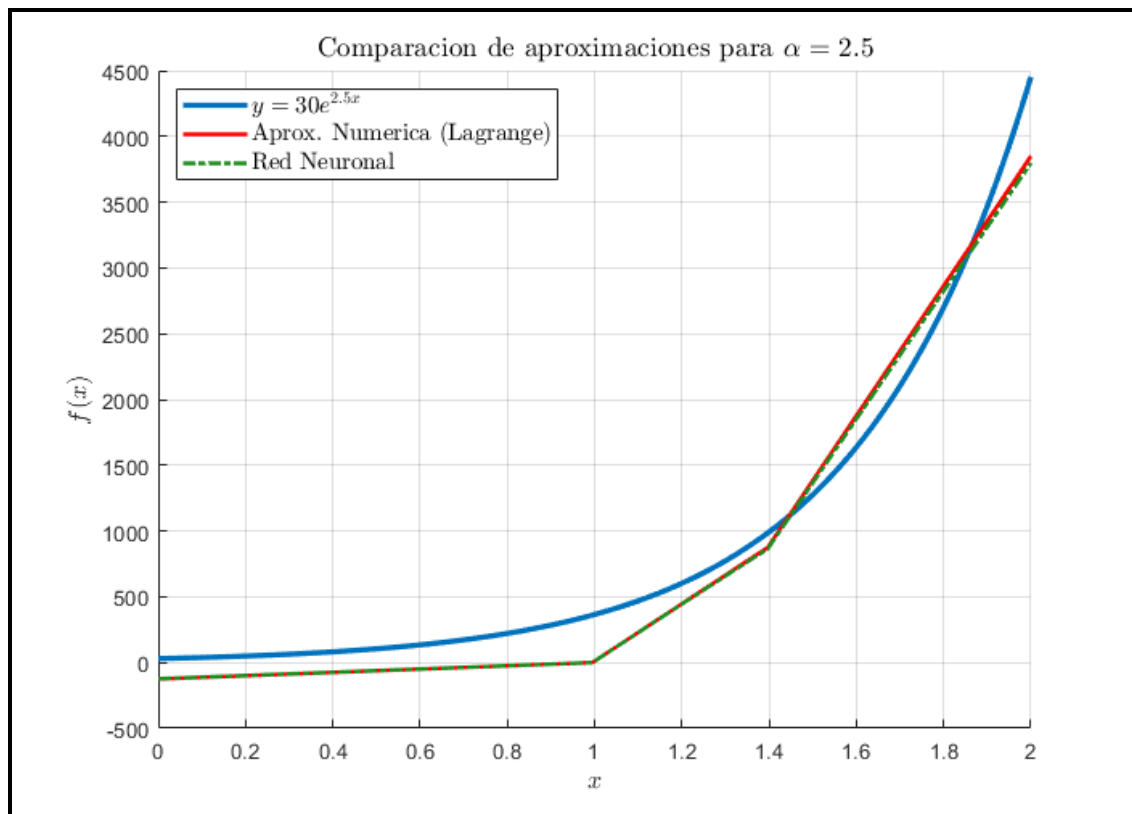


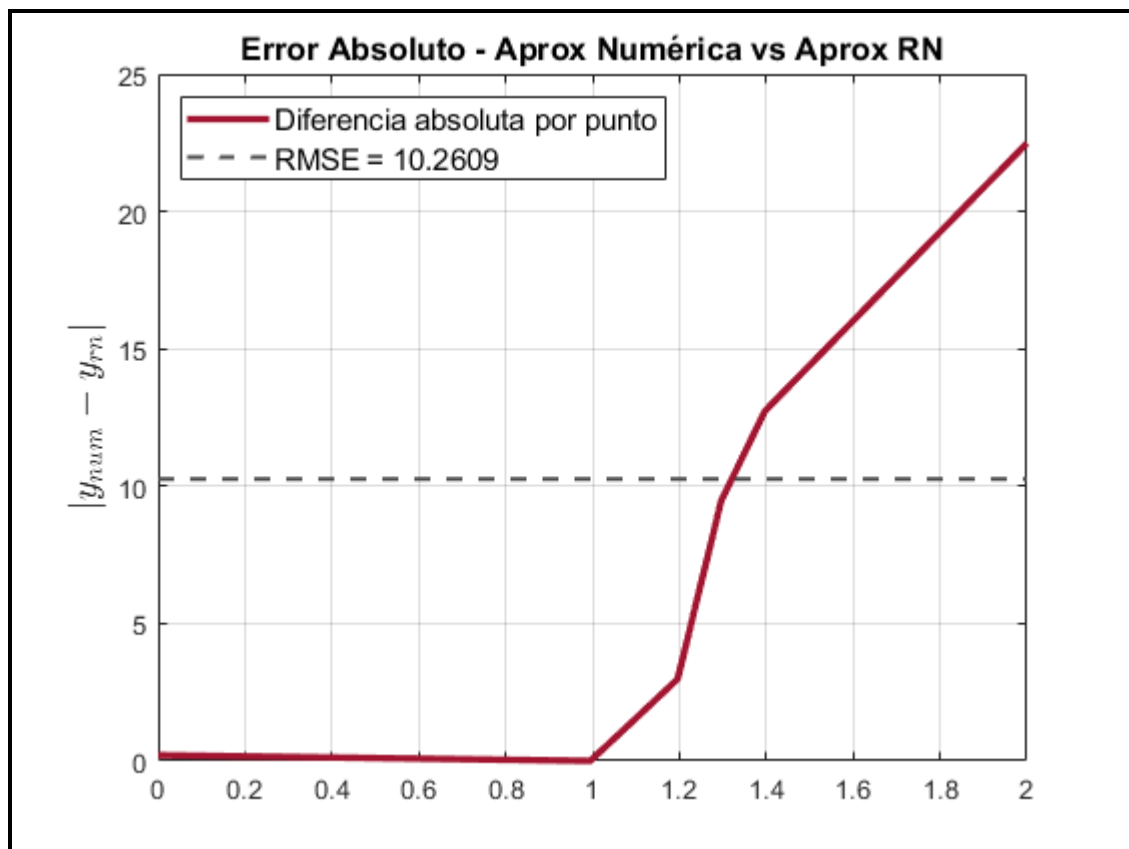
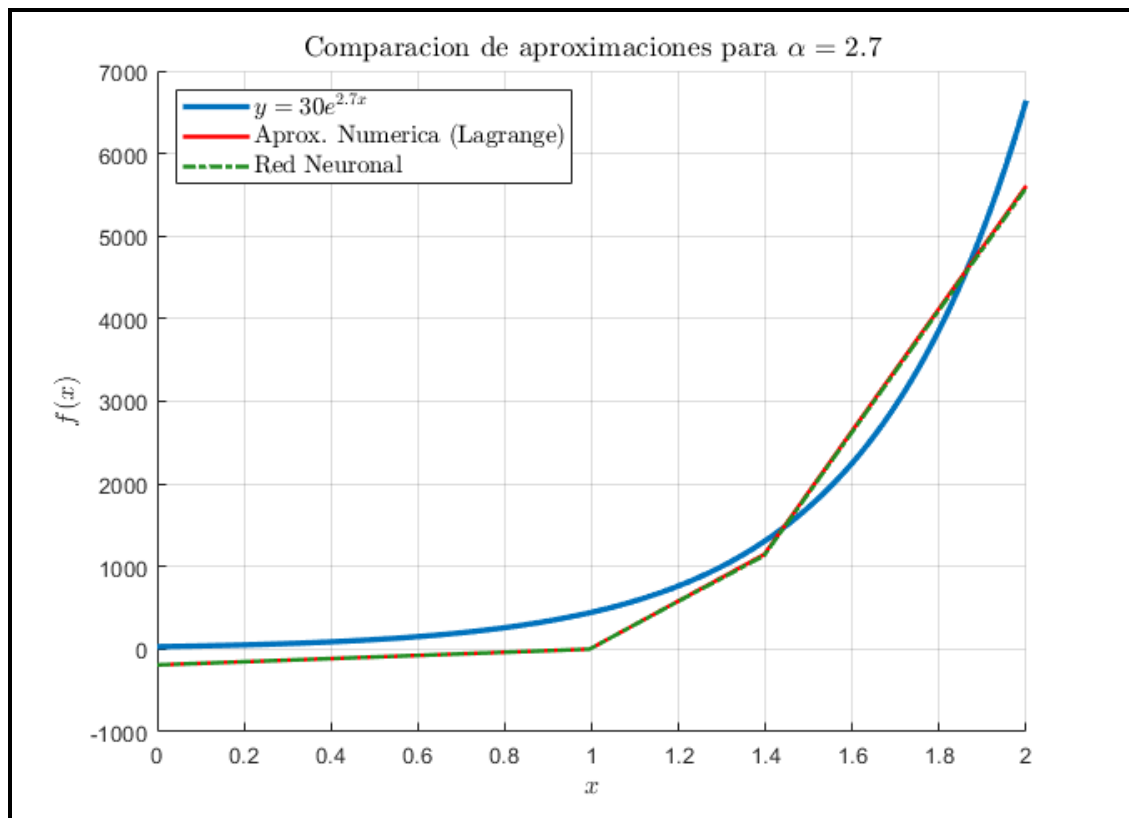


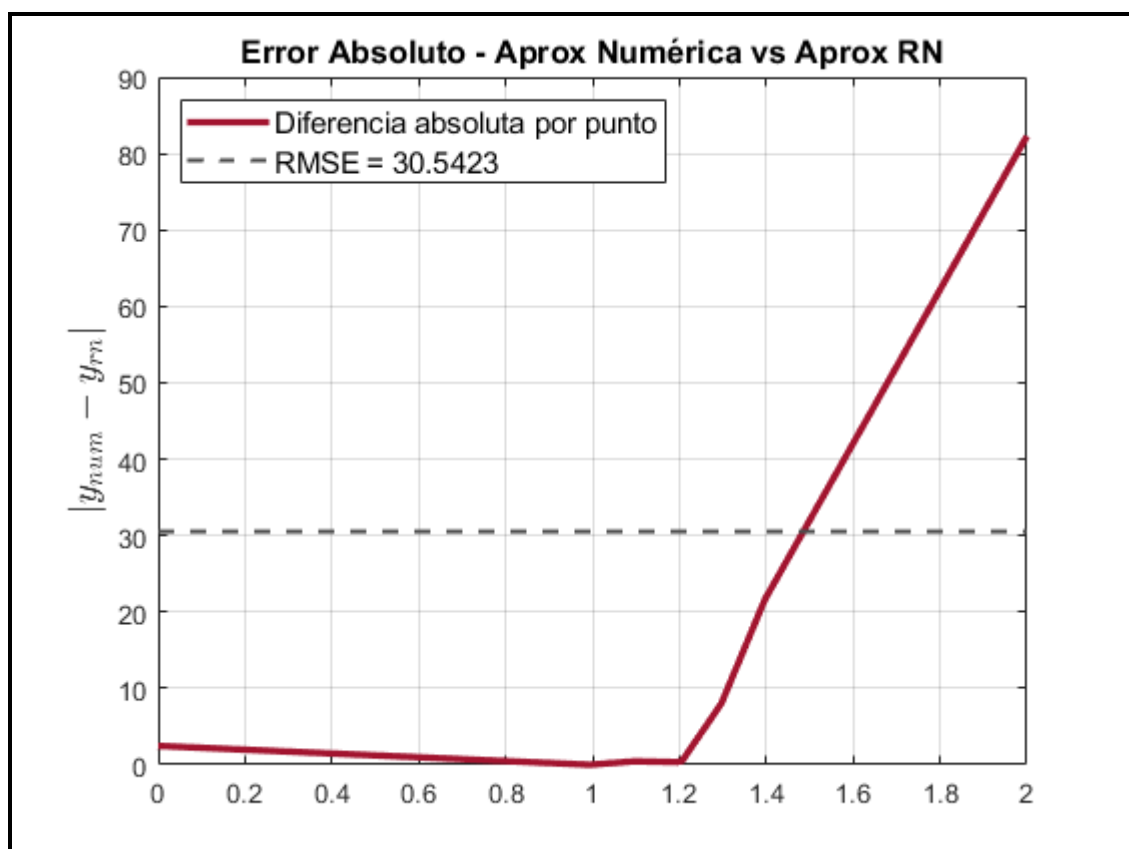
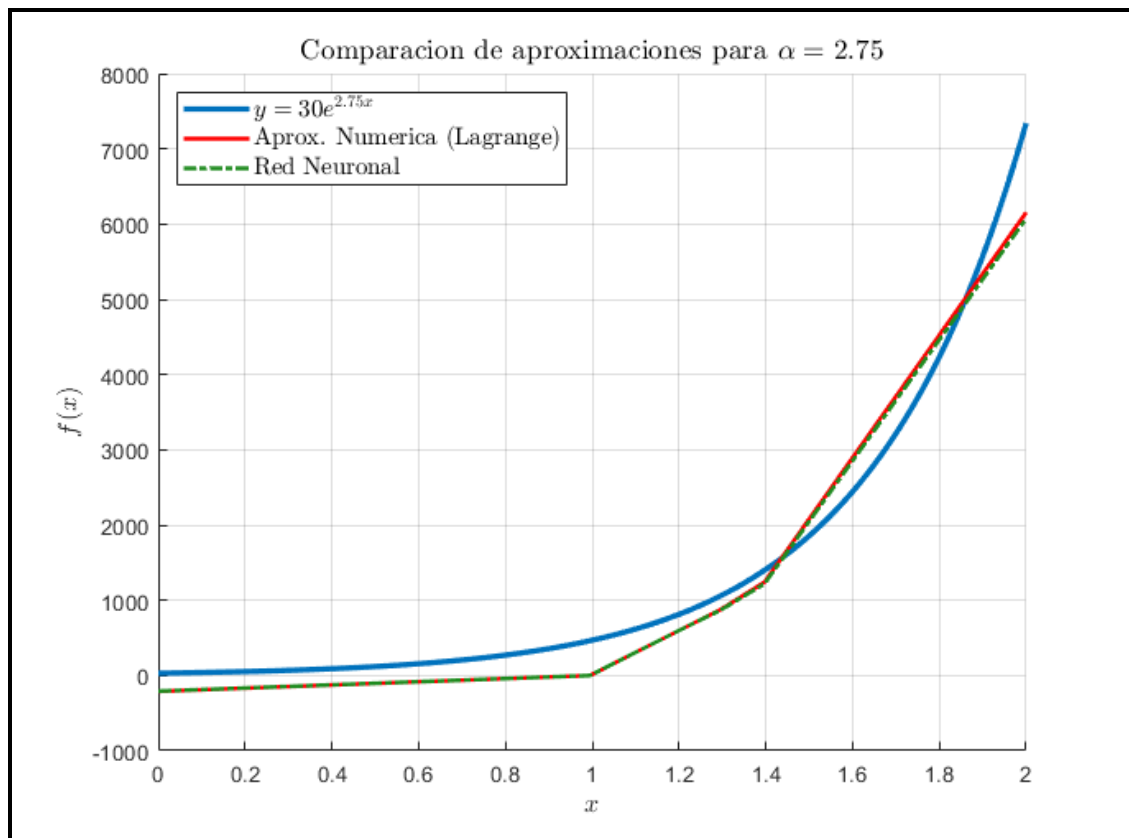


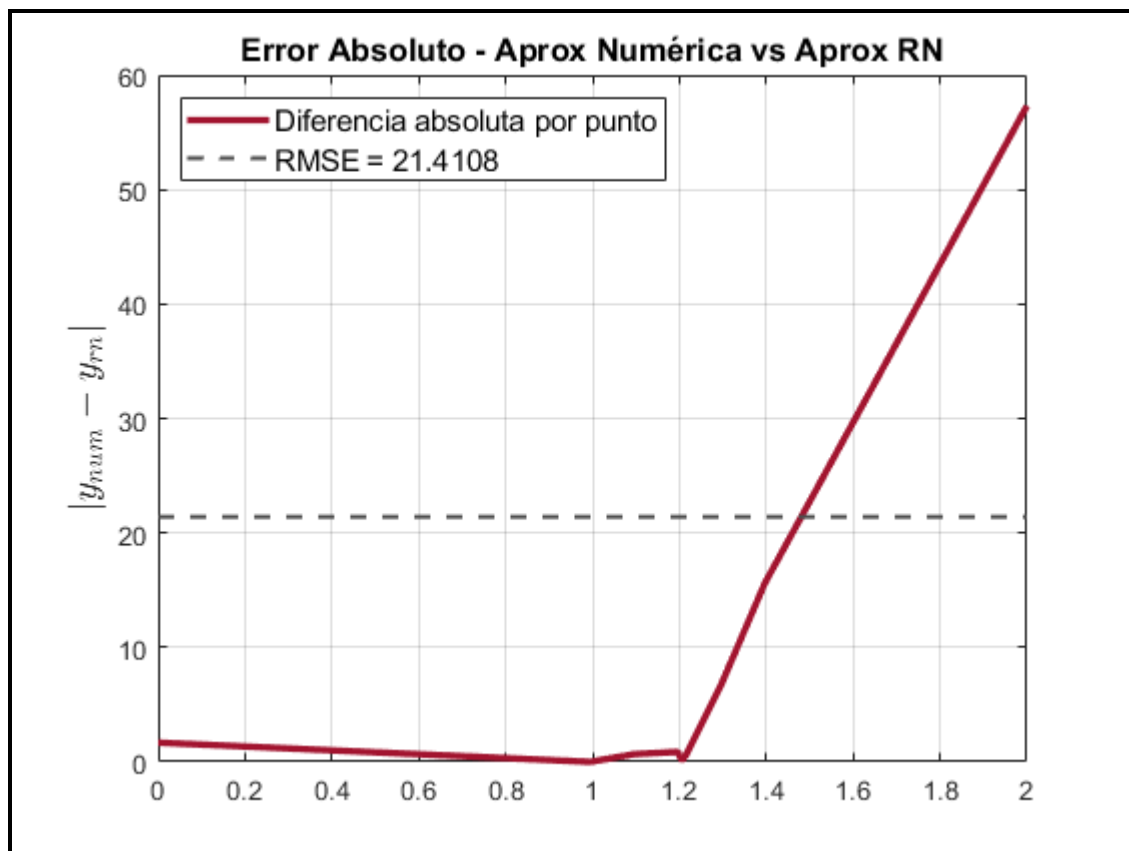
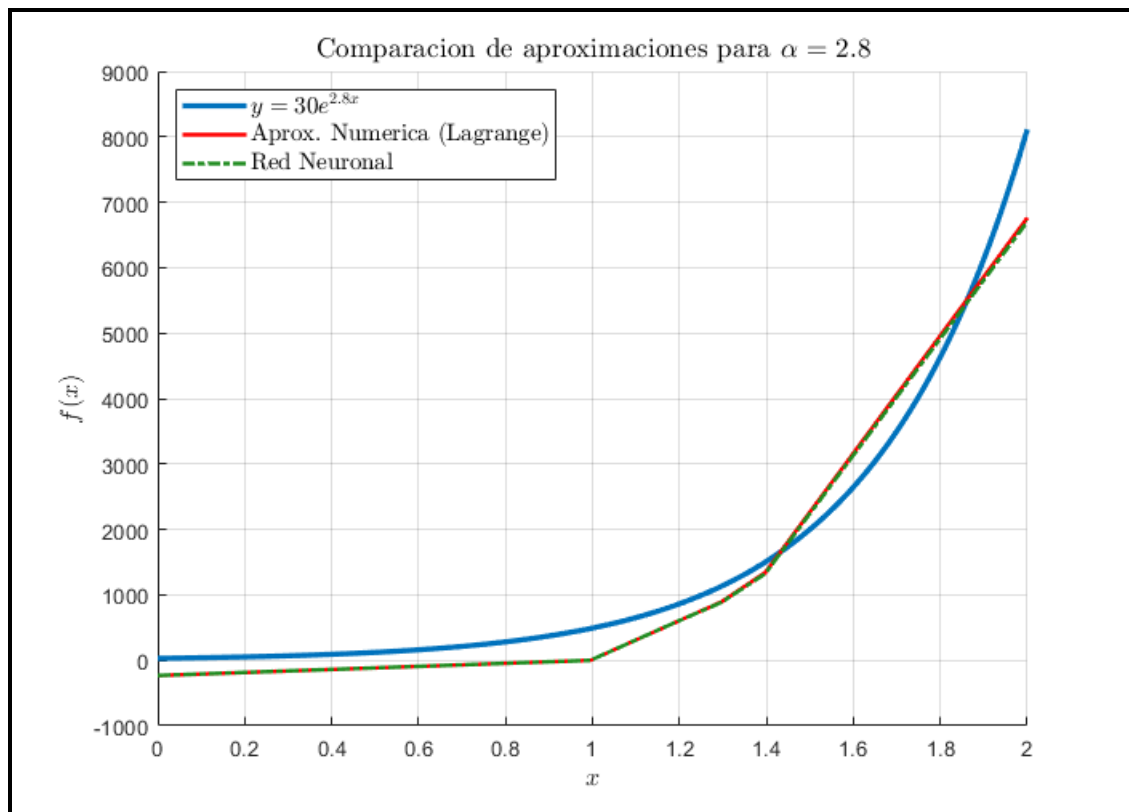


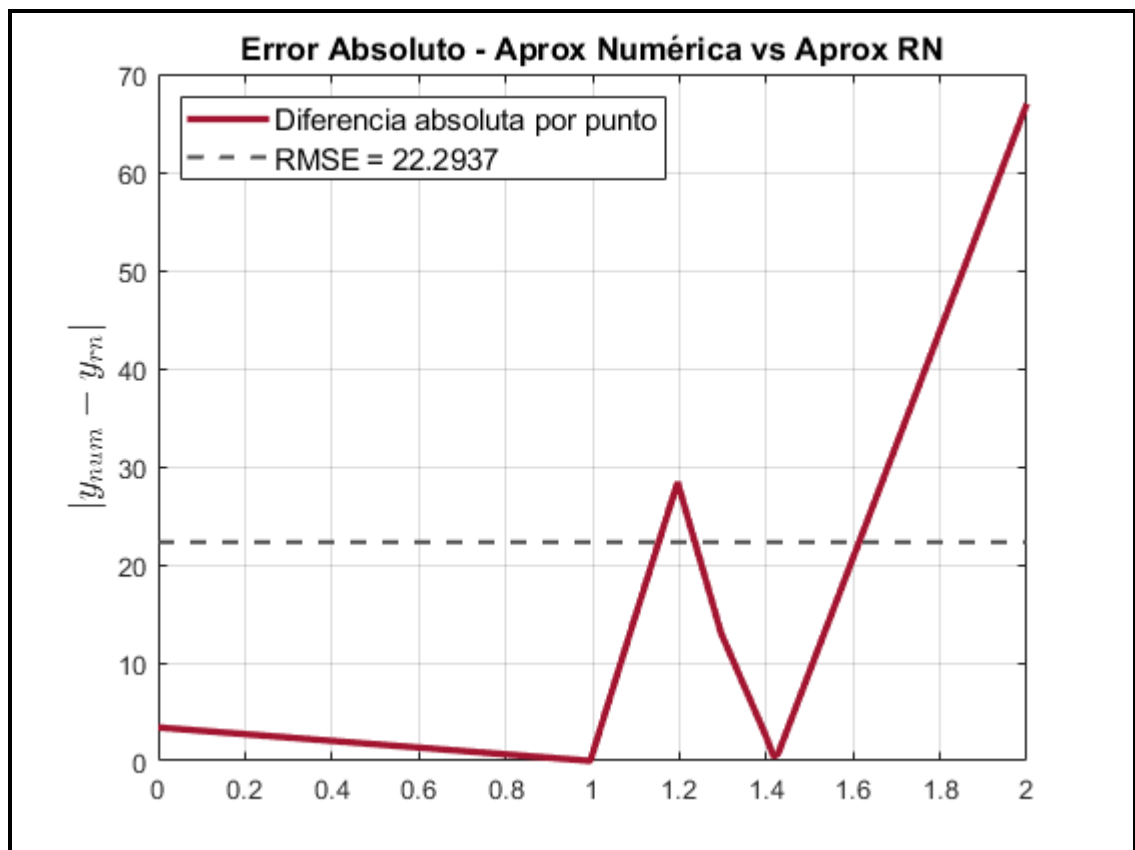
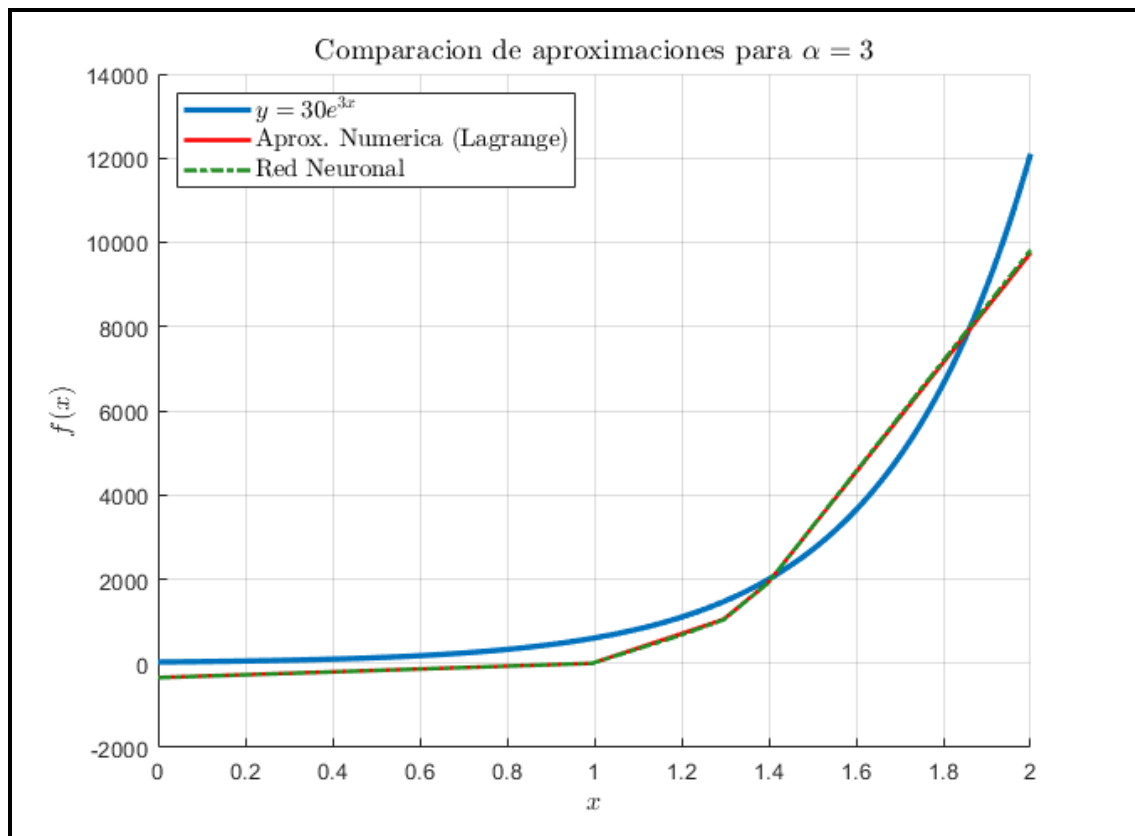


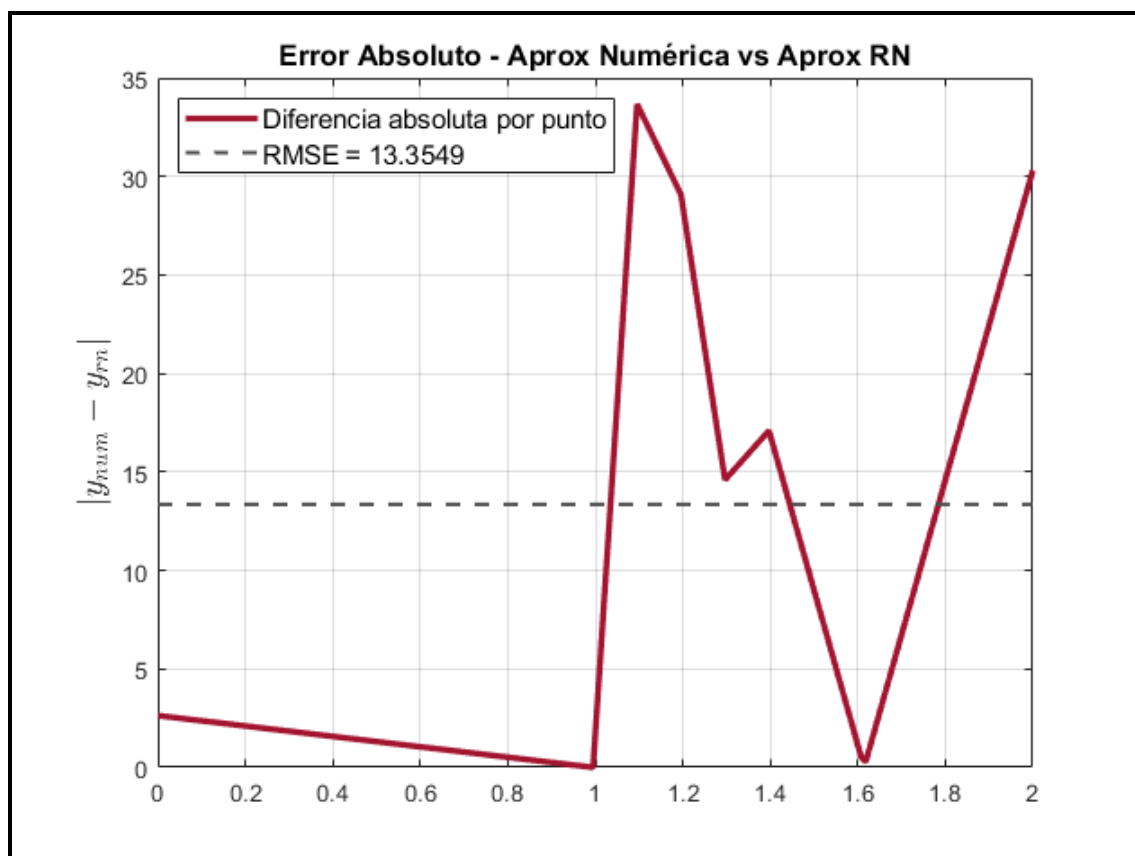
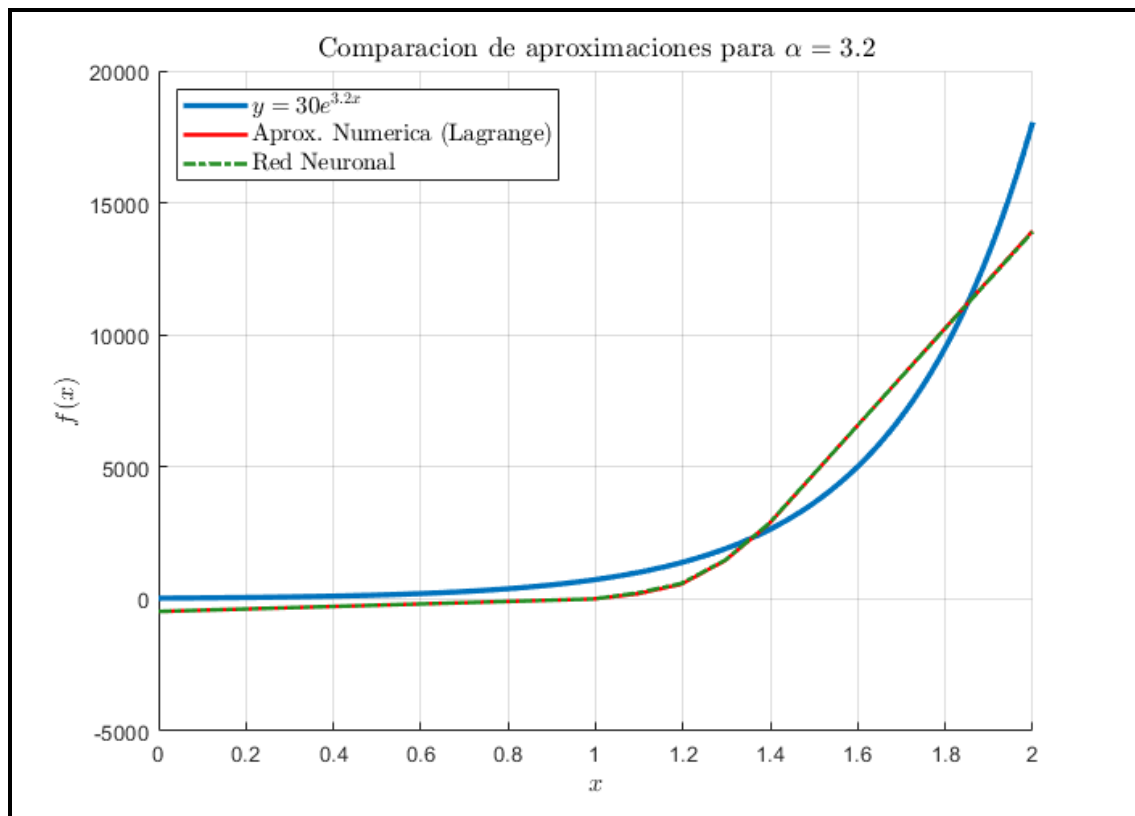


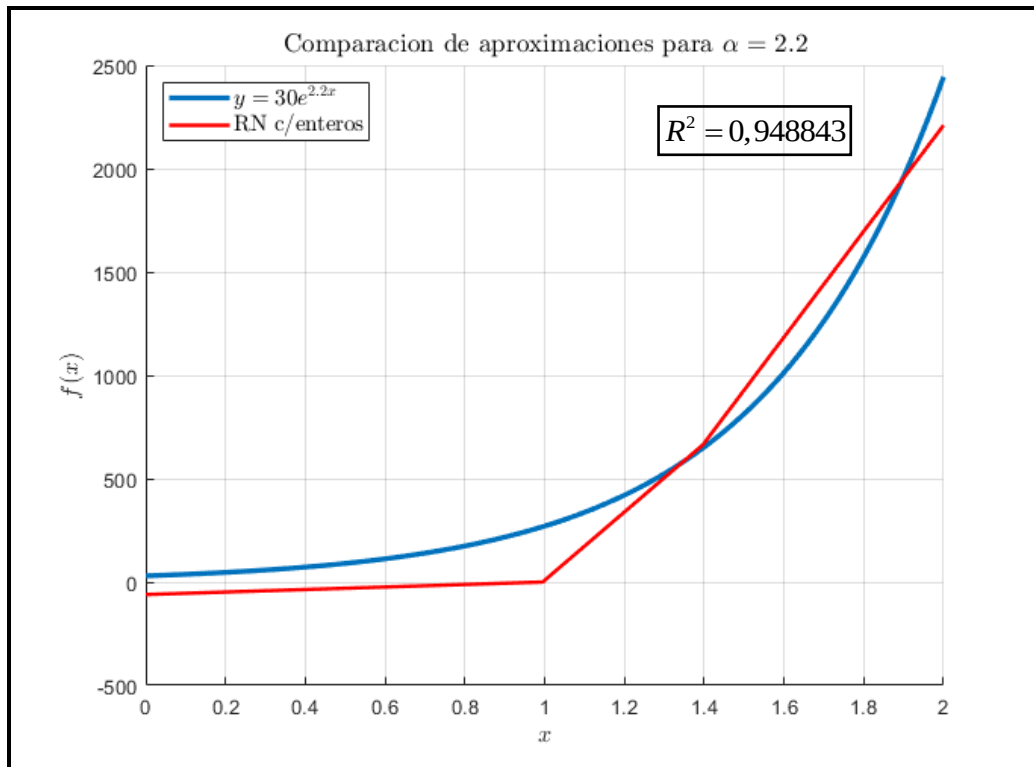




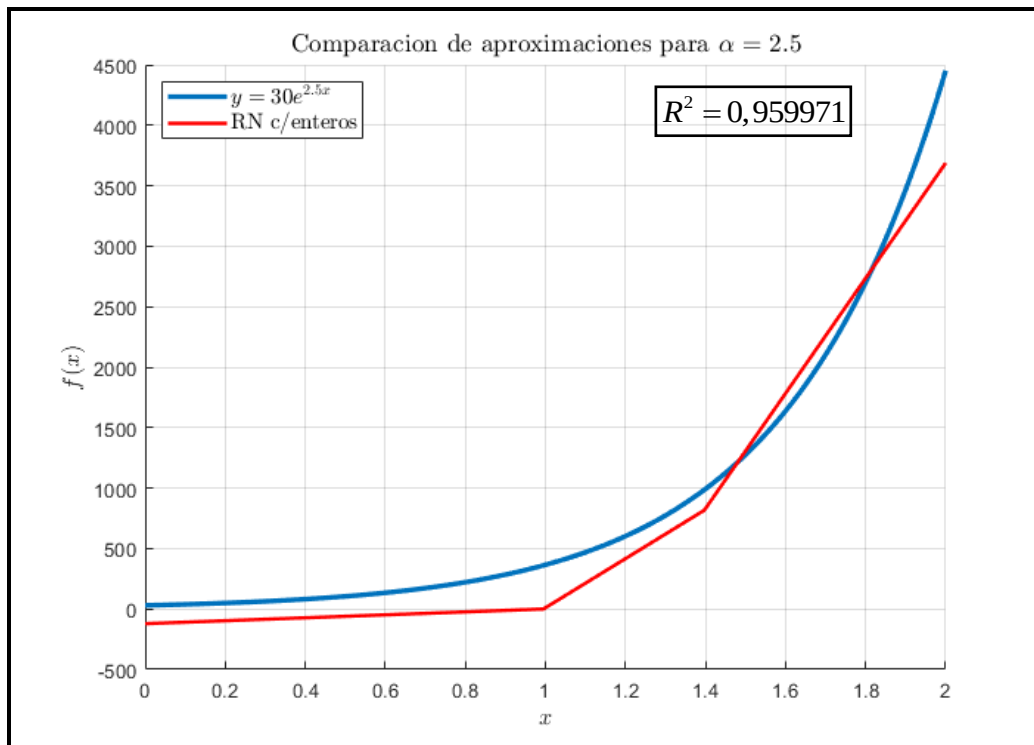




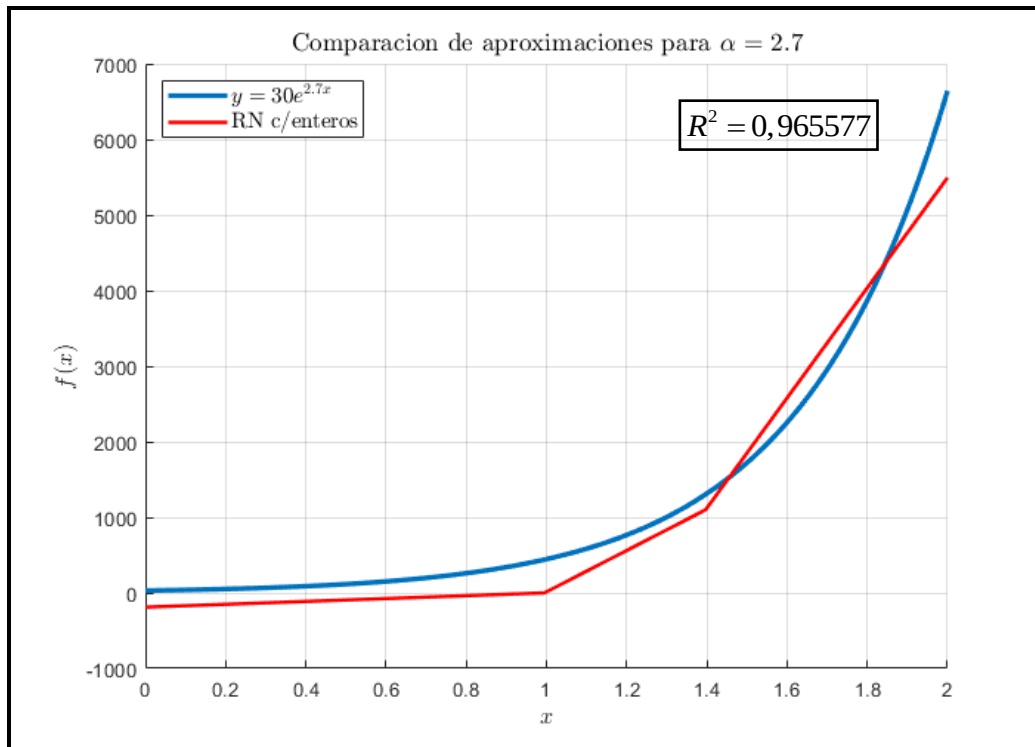




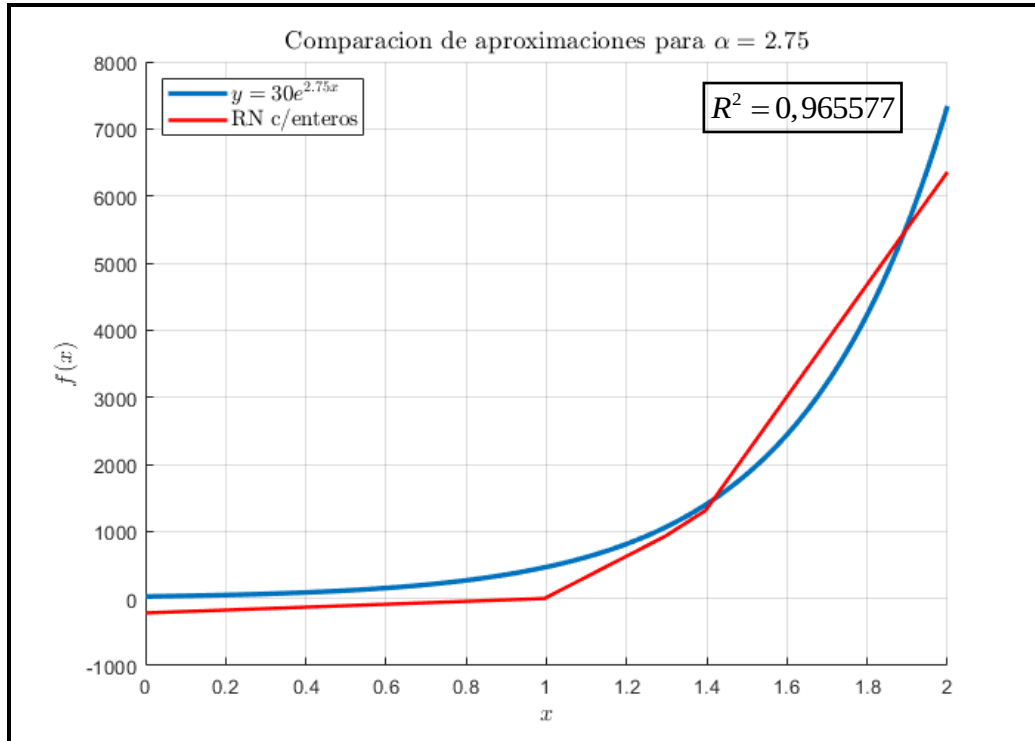
$$y_{aprox} = 5 \cdot b_1(x) + 0 \cdot b_2(x) + 0 \cdot b_3(x) + 0 \cdot b_4(x) + 4 \cdot b_5(x)$$



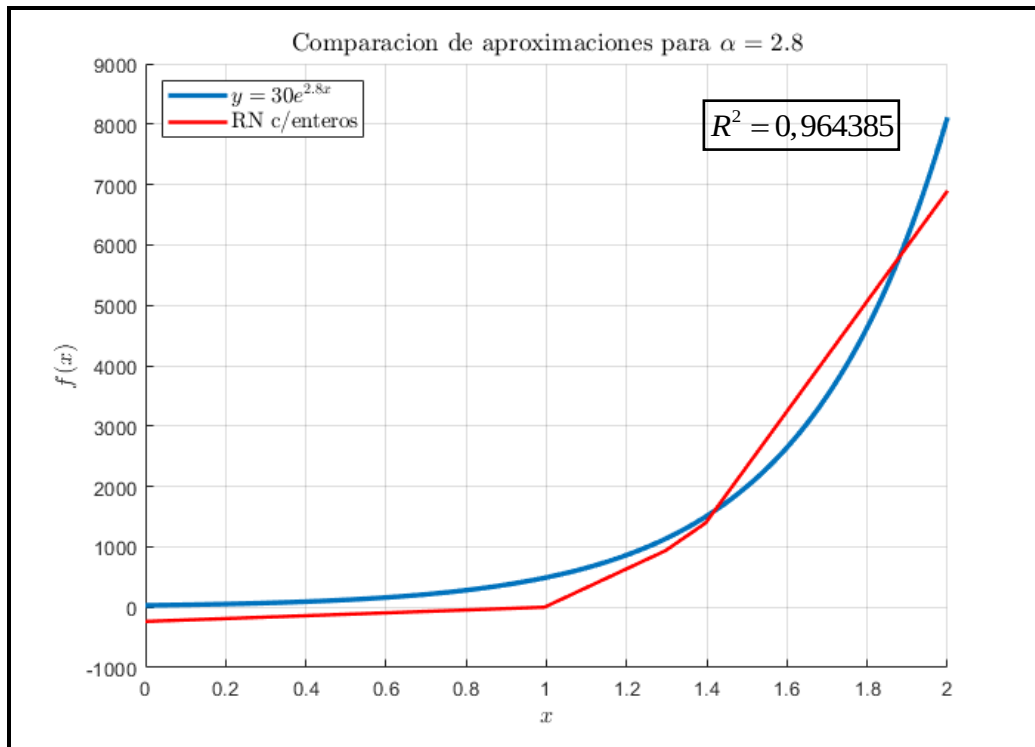
$$y_{aprox} = 6 \cdot b_1(x) + 0 \cdot b_2(x) + 0 \cdot b_3(x) + 0 \cdot b_4(x) + 12 \cdot b_5(x)$$



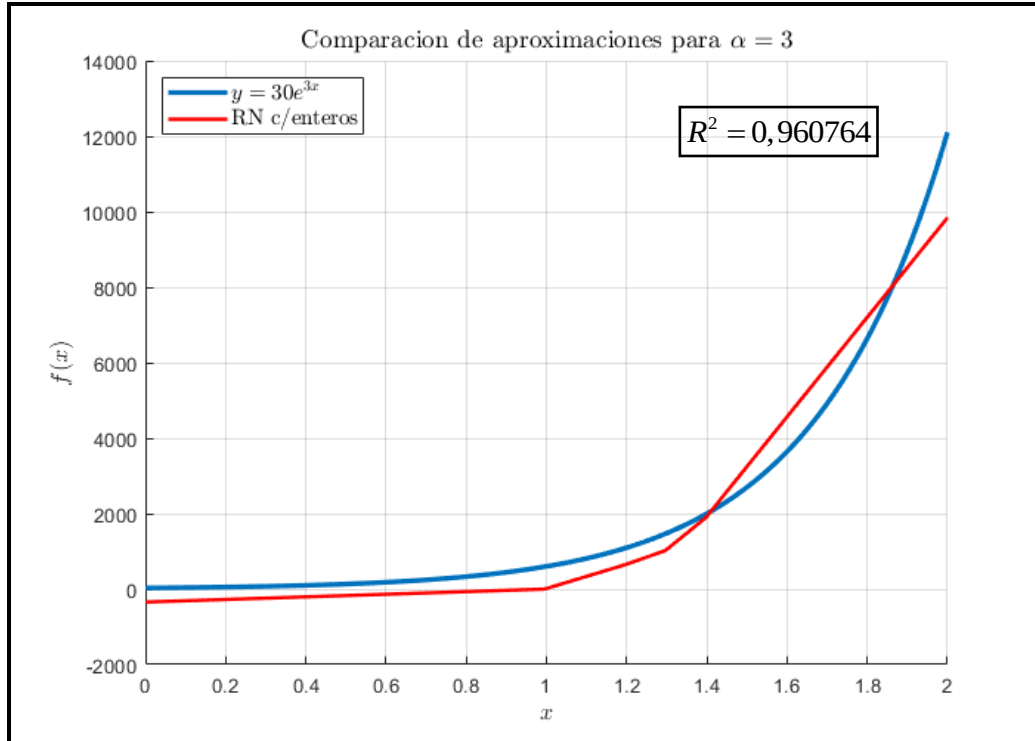
$$y_{aprox} = 8 \cdot b_1(x) + 0 \cdot b_2(x) + 0 \cdot b_3(x) + 0 \cdot b_4(x) + 20 \cdot b_5(x)$$



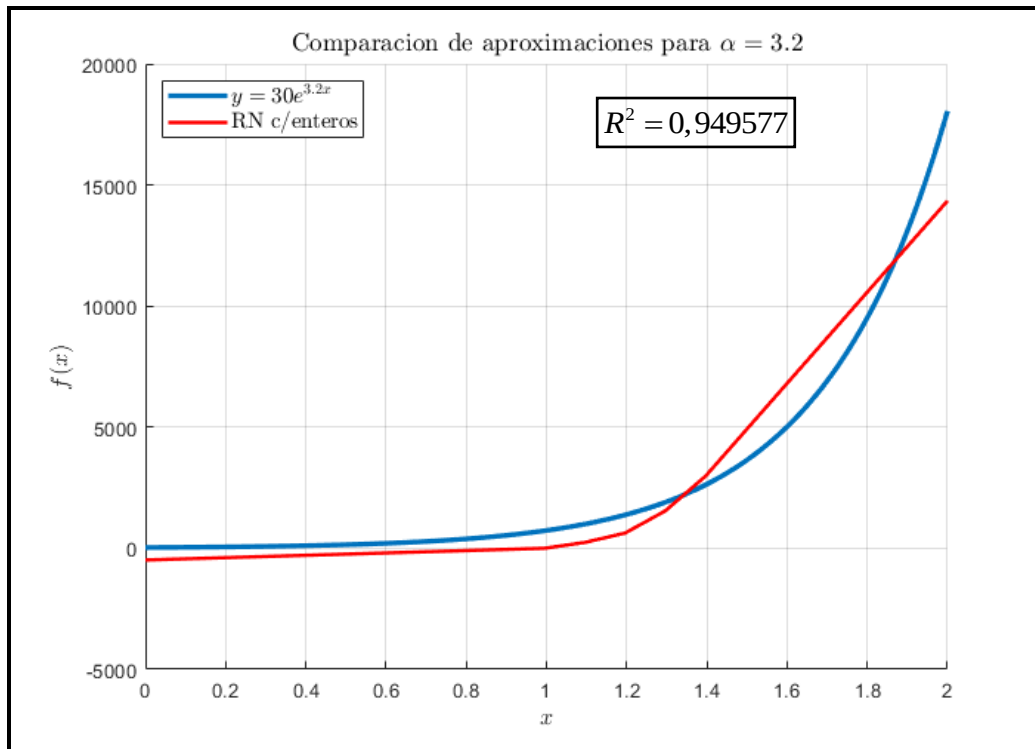
$$y_{aprox} = 9 \cdot b_1(x) + 0 \cdot b_2(x) + 0 \cdot b_3(x) + 3 \cdot b_4(x) + 20 \cdot b_5(x)$$



$$y_{aprox} = 9 \cdot b_1(x) + 0 \cdot b_2(x) + 0 \cdot b_3(x) + 6 \cdot b_4(x) + 20 \cdot b_5(x)$$



$$y_{aprox} = 9 \cdot b_1(x) + 0 \cdot b_2(x) + 2 \cdot b_3(x) + 20 \cdot b_4(x) + 20 \cdot b_5(x)$$



$$y_{aprox} = 6 \cdot b_1(x) + 5 \cdot b_2(x) + 20 \cdot b_3(x) + 21 \cdot b_4(x) + 20 \cdot b_5(x)$$

"El problema de optimización planteado consiste en minimizar la función cuadrática convexa $\|b - Ax\|^2$ sujeta a restricciones de caja $0 \leq x \leq 20$. Este es un problema de Programación Cuadrática (QP) en el caso de coeficientes reales, y de Programación Cuadrática Mixta-Entera (MIQP) cuando se impone la condición de integralidad. Para resolverlo, se empleó el método de los Multiplicadores de Lagrange (trust-constr) en el caso continuo, y una estrategia de búsqueda local en el caso entero."